GLOBAL INDUSTRIAL ARCHITECTURE

TIWAG POWER STATION CONTROL CENTRE

BY: NOOR ALKINANY

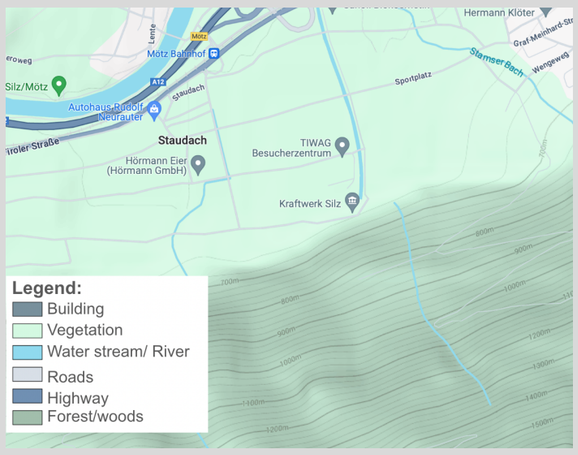
OVERVIEW

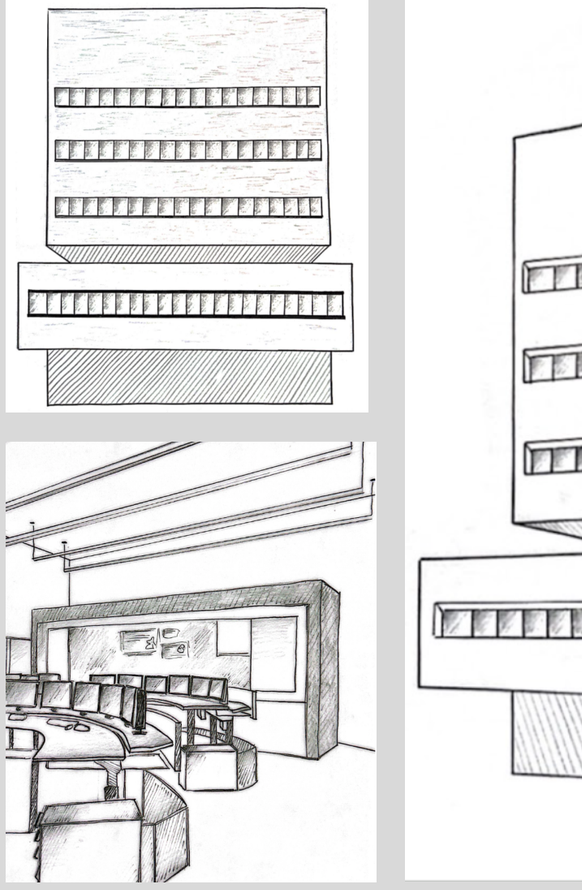
Location: Silz, Austria
Address: Doktor Meinrad Praxmarer Straße, 6424 Silz, Austria
Architects: Bechter Zaffignani Architekten
Building type: Industrial (Energy Plant)
Year: 2014

The power station control centre stands as an impressive tower like structure, purpose built for operational efficiency and functionality. Its design concept revolves around a vertical stacking of functional units, with each floor dedicated to a specific aspect of control and management. This hierarchal management not only enhances operational clarity but also facilitates the seamless administration of access and security protocols. By organising different functions on separate levels, the control centre optimises workflow and resource allocation. This vertical layout promotes an intuitive navigation of the facility, enabling personnel to swiftly move between departments and tasks. In essence, the control centre exemplifies a forward-thinking approach to infrastructure design, where form meets function to optimise performance and operational effectiveness in the vital field of power generation and distribution.

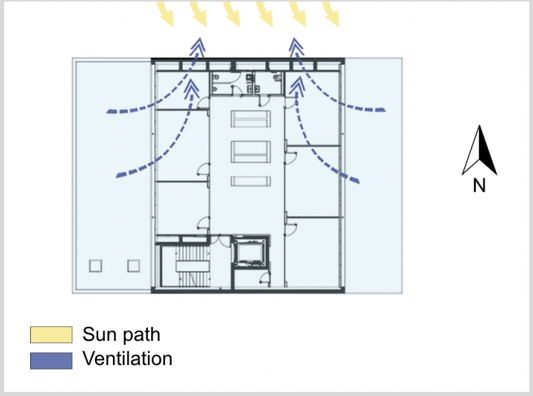
ANALYSIS

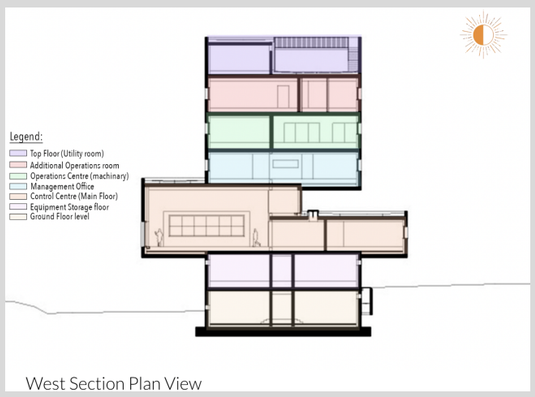
SITE ANALYSIS





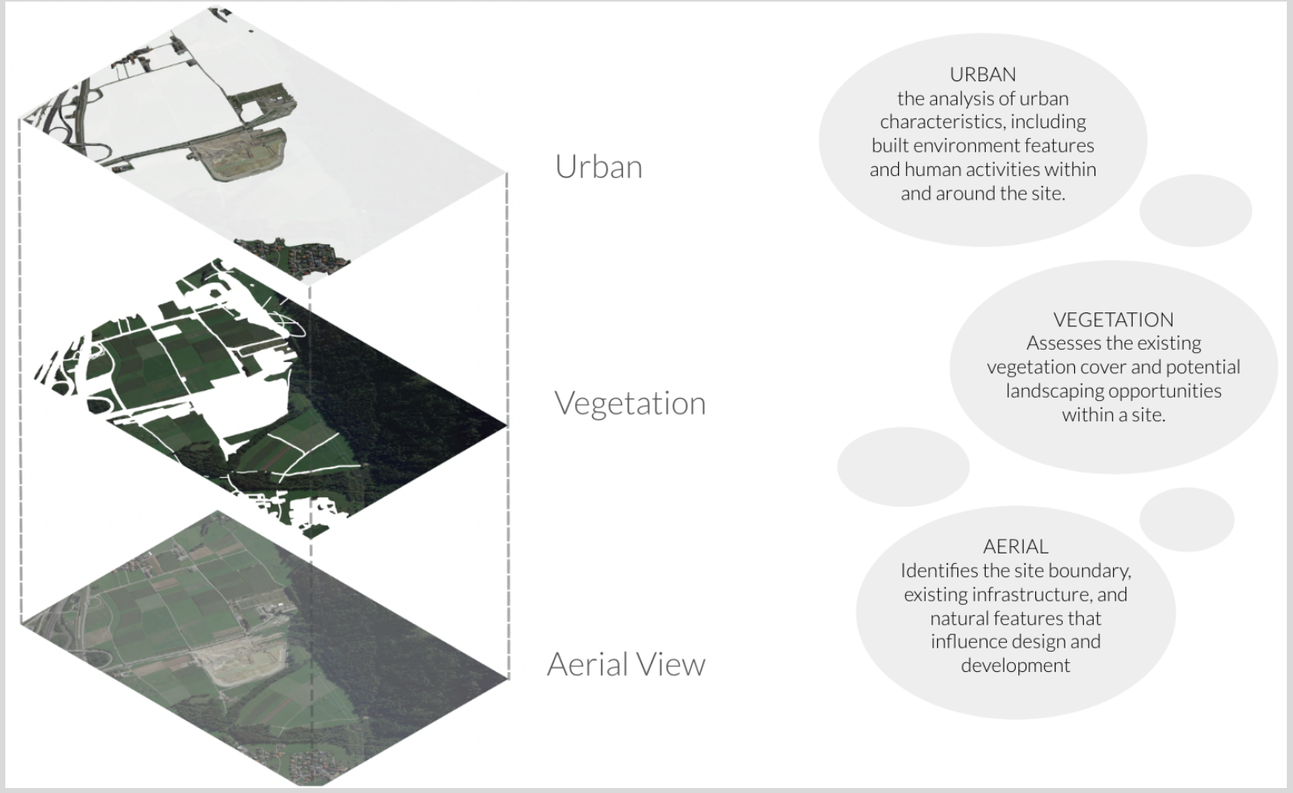
FLOOR PLAN ANALYSIS



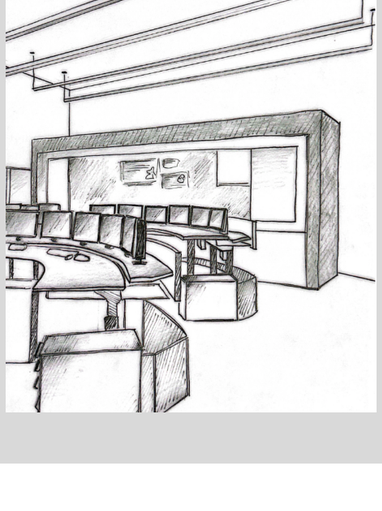


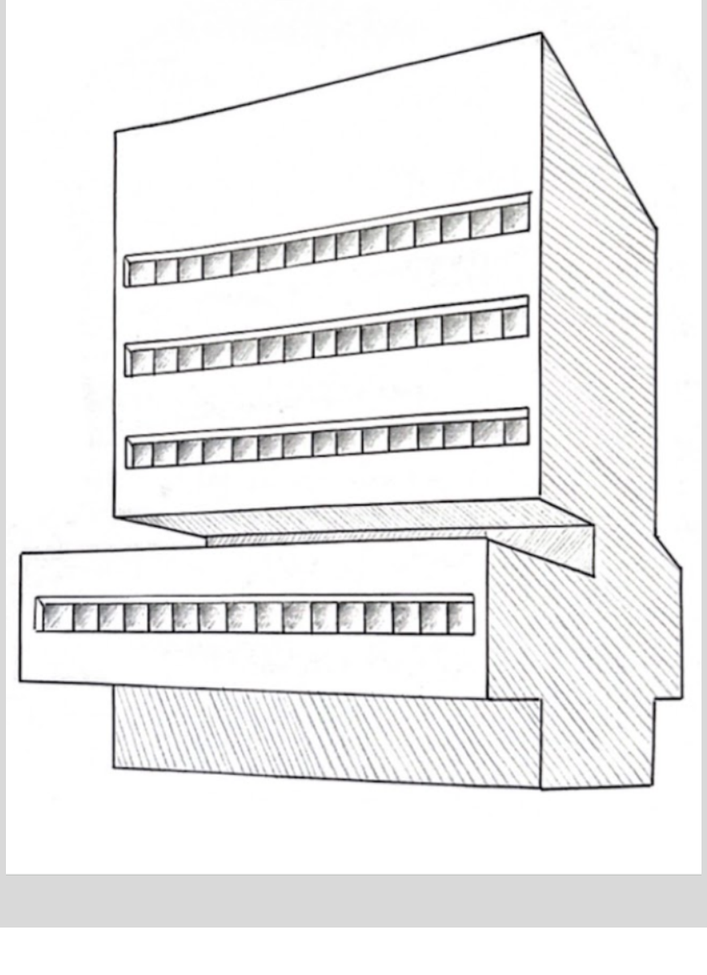
West Section Plan View

3D SITE ANALYSIS DIAGRAM

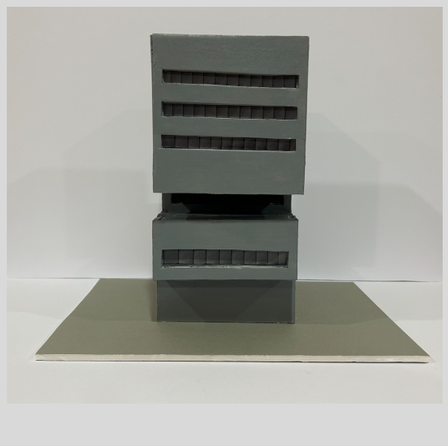


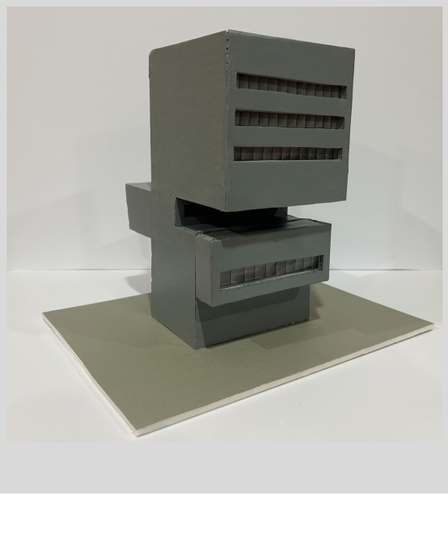
SKETCHES

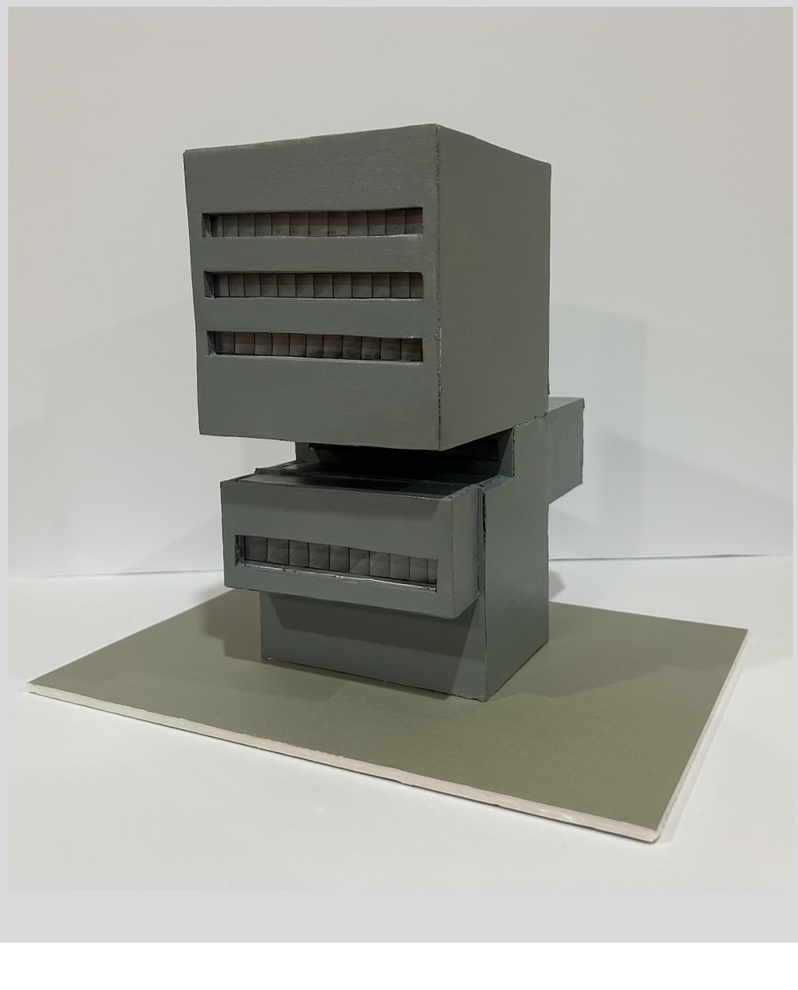




MODEL







GLOBAL INDUSTRIAL ARCHITECTURE

NATIONAL ARCHIVES PRESERVATION FACILITY

BY: MAISIE RICHTER

OVERVIEW

Choosing Global Industrial Architecture was the focus of our project, a decision fueled by a blend of personal interests and academic aspirations within our group (Maisie, Ornela, Sam and Noor).

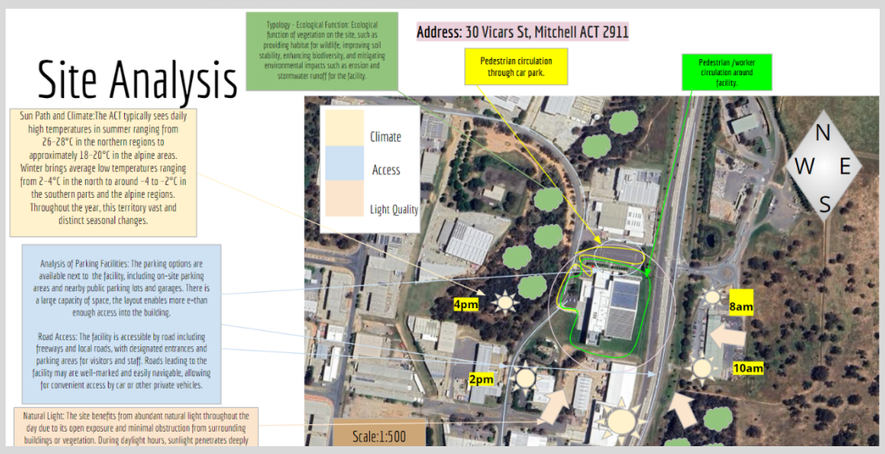
Through the extensive exploration of the National Archives preservation facility within the context of global industrial architecture, I've gained invaluable insights into the interplay among historical preservation, architectural design, and industrial functionality. This journey deepened my understanding of how built environments intersect with cultural significance worldwide. Examining the facility emphasized the critical significance of adaptive reuse and sustainable design principles in revitalizing industrial spaces. From scrutinizing its vertical section to analyzing spatial layout and environmental considerations, I recognized the multifaceted approach needed to balance preservation goals, functionality, and user needs. Delving into global industrial architecture expanded my perspective on diverse architectural styles and construction methods employed worldwide. This study highlighted industrial architecture's role as a tangible embodiment of cultural heritage and technological progress, shaping community and national identities. Ultimately, this journey underscored the importance of interdisciplinary collaboration and holistic problem-solving in architectural preservation and industrial heritage. By integrating insights from architecture, engineering, history, and cultural studies, innovative solutions can be developed that honor the past while embracing the future. Inspired by this experience, I aim to contribute meaningfully to the preservation and advancement of our built environment globally.

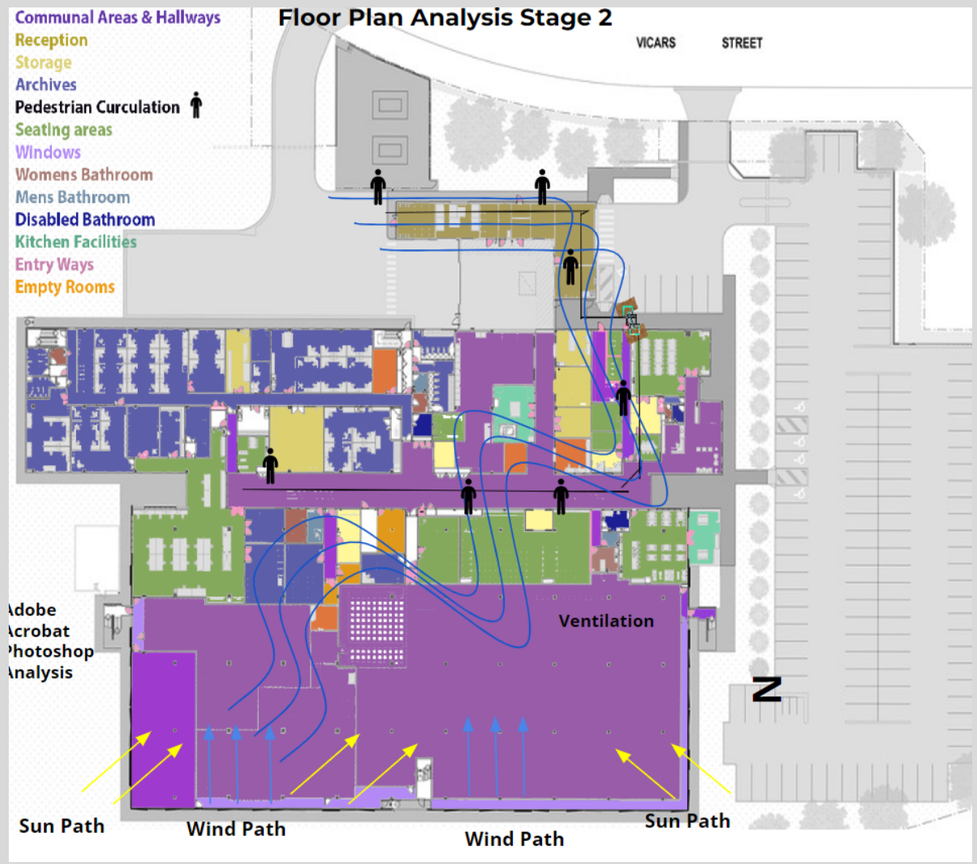
Architects: May + Russell Architects
Area: 18000 m²
Built Year: 2017
Address: 30 Vicars St, Mitchell ACT 2911
Building Type: Warehouse

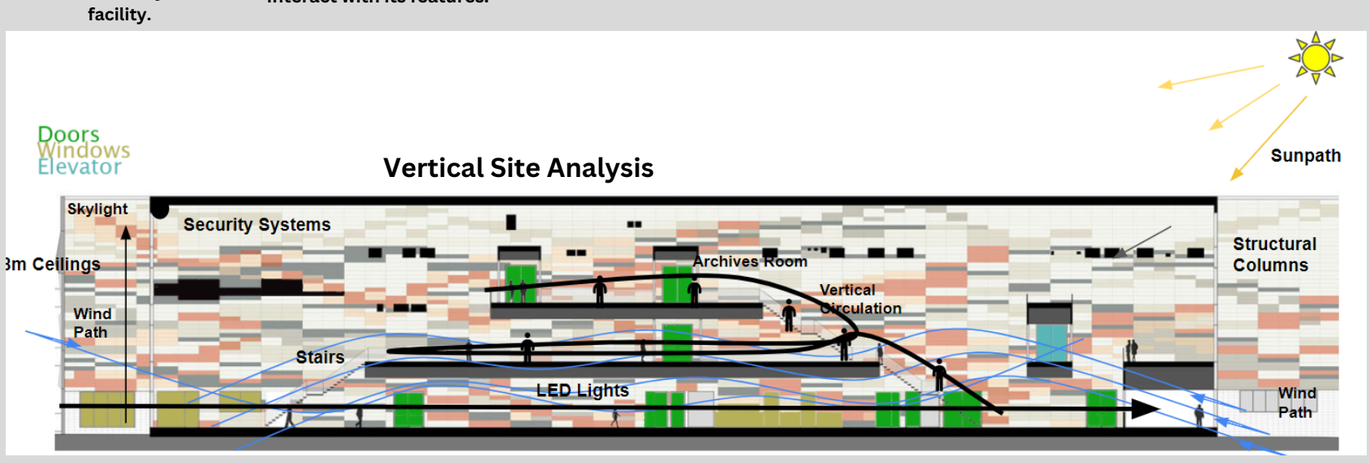
ANALYSIS

My journey commenced with a meticulous examination of the site surrounding the National Archives Preservation Facility. Through site analysis, I discerned not only the physical characteristics but also the historical and cultural context that informed the facility's design and location. By scrutinizing factors such as topography, orientation, and surrounding infrastructure, I gained invaluable insights into how the site's features influenced the architectural decisions and functionality of the facility.

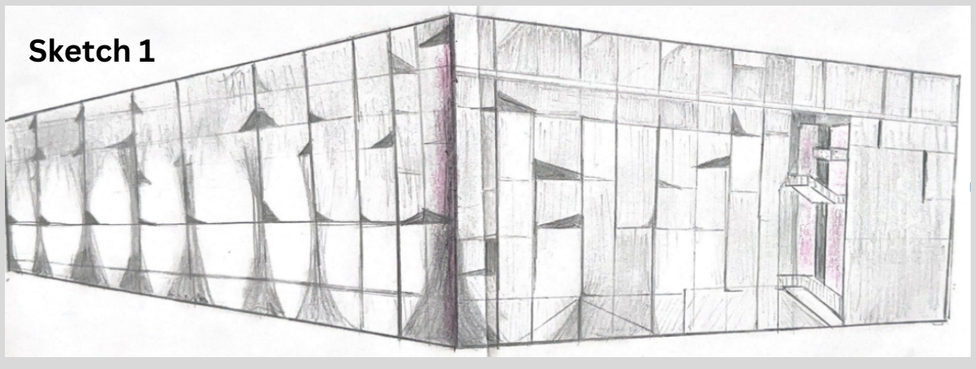
Moving further, my floor plan analysis delved into the intricate details of the floor plan, unraveling the spatial layout and functional organization of the facility. With a keen eye for detail, I dissected each room and space, deciphering their intended purposes and interrelationships. The floor plan served as a blueprint for understanding circulation patterns, access points, and the integration of structural elements, guiding me in comprehending how users navigate the space and interact with its features.

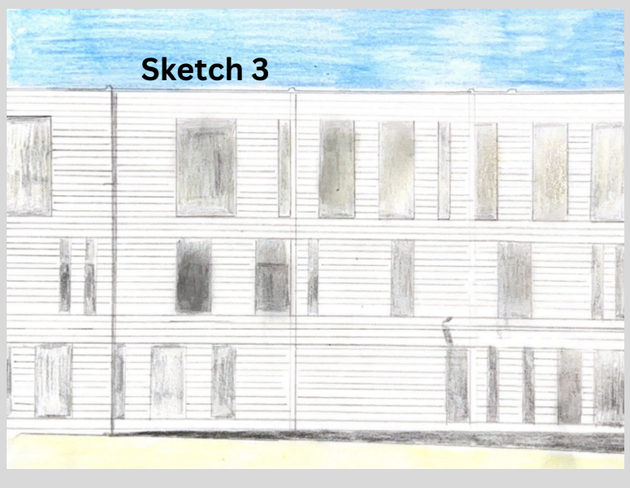




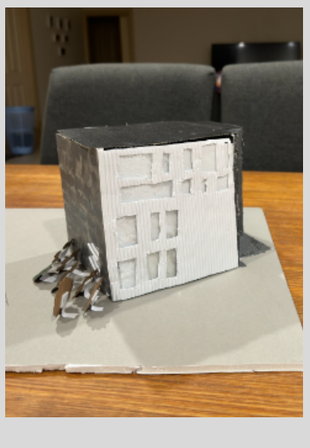


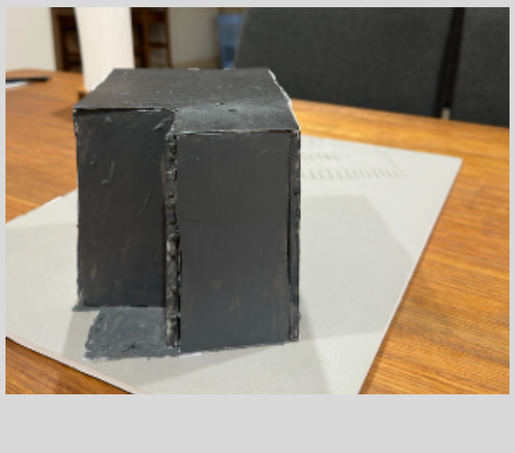
SKETCHES

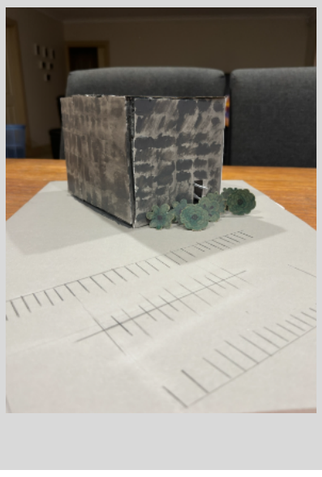




FINAL MODEL







GLOBAL INDUSTRIAL ARCHITECTURE

BREWING FACTORY

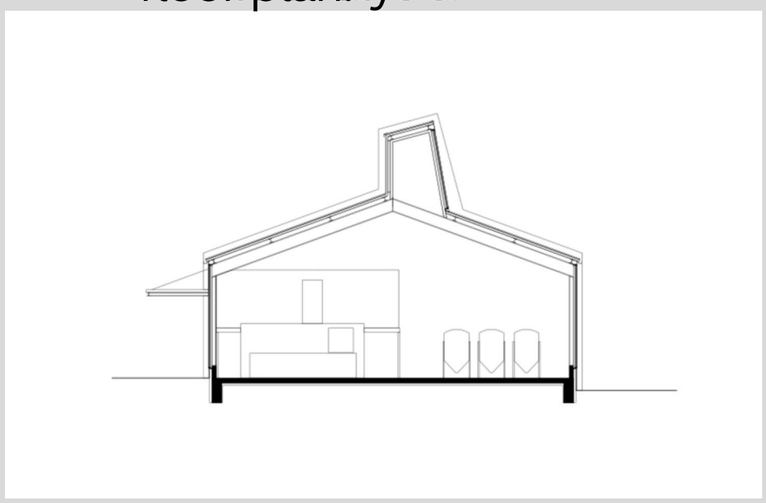
BY: SAM IBRAHIM

OVERVIEW

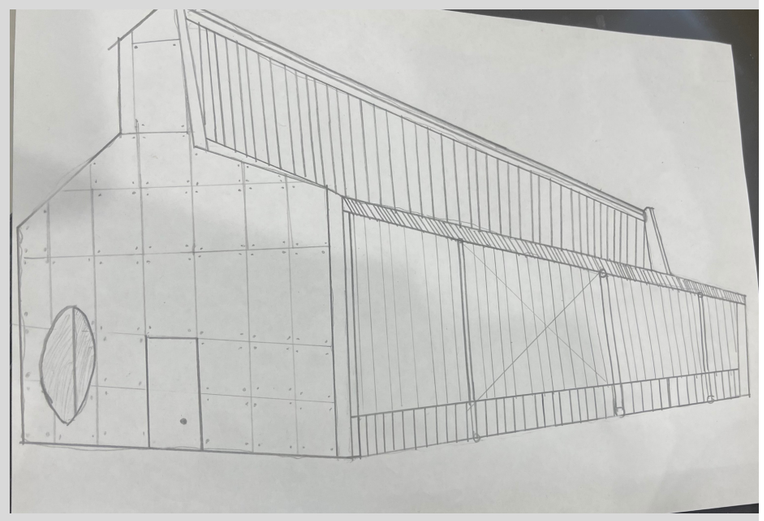
A brewing factory is a facility that specializes in the production of various types of beer and other alcoholic beverages. These factories typically consist of a series of large tanks and machinery used to ferment and package the beverages.

ANALYSIS

site plan analysis:



floor plan:



perspective drawing:t

MODEL

GLOBAL INDUSTRIAL ARCHITECTURE

BERTOLOTTI VIAL MACHINERY AND OFFICE WAREHOUSE

BY: ORNELA SULCE

OVERVIEW

Bertolotti Vial Warehouse and Office
Santa Fe, Argentina
Barn/Warehouse
2019

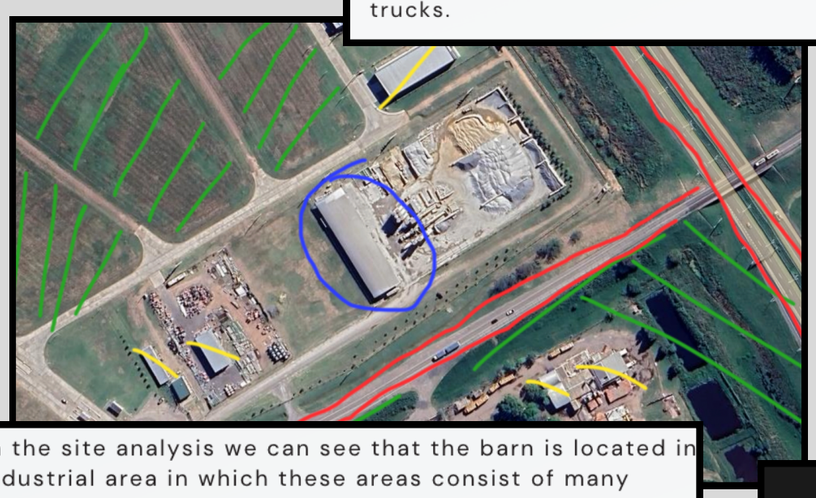
My personal interest led me to focus specifically on the Bertolotti Vial Machinery and Office Warehouse which is also classified as a barn. What drew me to this building was the very minimalistic appearance it had compared to many other industrial buildings. Aside from the appearance I was interested in the spatial aspect of the building in terms of how space is optimised in a way where the workflow is smooth while also ensuring safety. I also wanted to look into how technology is implemented and how they could be impacting productivity and possibly reshaping the industry. I found it's design to be quite unique for a warehouse which made it eye-catching for me.

One key takeaway from this design is how both aesthetics and functionality can be balanced when creating a design. The Bertolotti Vial Machinery and Office Warehouse integrates office spaces with machinery and warehouse functions which exemplifies how a building can still possess operational efficiency while also being visually appealing.

ANALYSIS

SITE ANALYSIS

- Another prominent feature of this site is the roads which are going in straight directions without too many turns. This indicates the kind of vehicles that are driven on those roads which are typically bigger vehicles such as trucks.



- From the site analysis we can see that the barn is located in an industrial area in which these areas consist of many warehouses, factories, distribution centres, etc. We can see the same features in this site analysis as the barn (blue) is surrounded by other industrial buildings.
- Alongside, there is a lot of greenery surrounding it which could indicate that there was elements of sustainability implemented during the design process of this building.

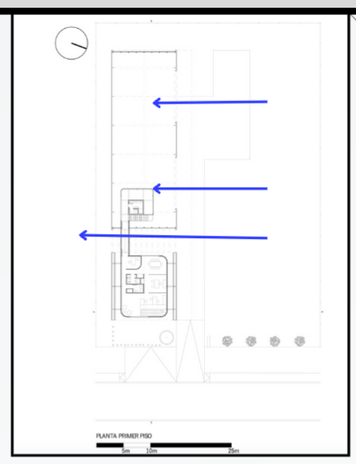
Legend

- Roads
- Greenery/Vegetation
- Other warehouses/barns
- Bertolotti warehouse

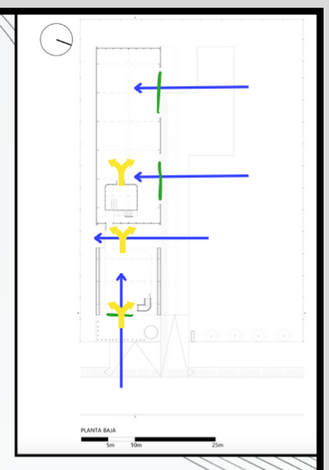
FLOOR PLAN ANALYSIS

- This building is a very well ventilated building due to it containing a lot of openings and entries (as shown with the blue arrows).
- The amount of opening allows for a lot of sunlight to flow throughout the whole building (as shown by the yellow arrows).

LEGEND
- Ventilation
- Sunlight
- Entry/Openings

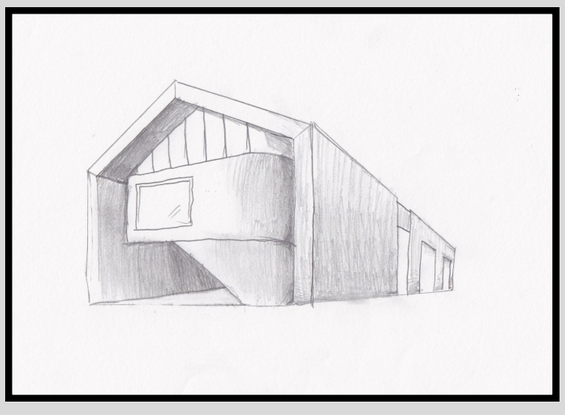


First floor

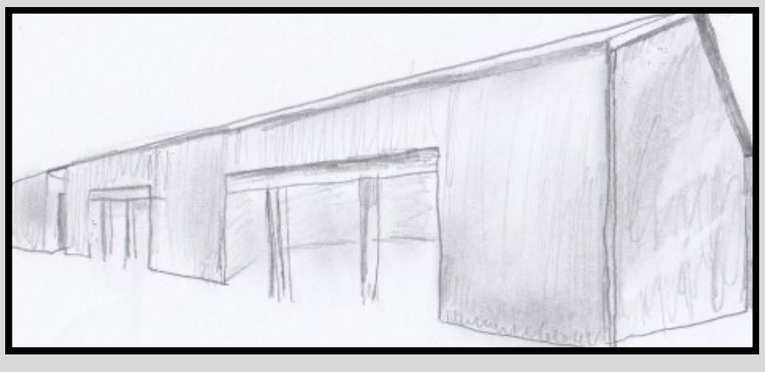


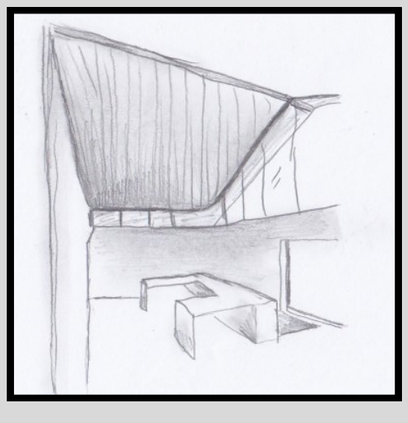
Ground floor

MODEL



PERSPECTIVE DRAWINGS





PHYSICAL MODEL